← Back **Quiz for Week 4**
Graded Assignment • 20 min

Due Sep 5, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. (**Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file here. If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

1 / 1 point

A retailer is importing a product from an overseas supplier. If there are q units on the market, the unit price of the product will be $a - bq$ dollars (this is called the *market clearing price*. The unit procurement cost of the product is c. The retailer wants to determine a procurement quantity that maximizes its profit. In an NLP that maximizes the retailer's profit, which of the following is the correct objective function?

- ( ) $\max\ q(aq - b - c)$.
- (•) $\max\ q(a - bq - c)$.
- ( ) $\max\ q(a - b - cq)$.
- ( ) $\max\ (q - c)(a - bq)$.
- ( ) None of the above.

✓ Correct

2. To linearize the following nonlinear program

1 / 1 point

$$\begin{array}{ll} \max & 5x_1 + 3x_2 \\ \text{s.t.} & \max\{x_1, 12\} \leq \min\{16, x_1 + 2x_2\} \\ & x_1 + 4x_2 \leq 20 \\ & x_1 \geq 0, x_2 \geq 0, \end{array}$$

we should replace the first constraint by which of the following?

- (•) $x_1 \leq 16, 12 \leq x_1 + 2x_2$.
- ( ) $x_1 \leq 16, 12 \geq x_1 + 2x_2$.
- ( ) $x_1 \geq 16, 12 \leq x_1 + 2x_2$.
- ( ) $x_1 \geq 16, 12 \geq x_1 + 2x_2$.
- ( ) None of the above.

✓ Correct

3. The IEDO county is trying to determine where to place one fire station. The locations of the county's six major towns are given in the following coordinates measured in kilometers. City 1: $(20, 20)$; city 2: $(60, 10)$; city 3: $(40, 30)$; city 4: $(40, 60)$; city 5: $(30, 0)$; city 6: $(10, 70)$. City 1 has in average 20 fires per year. For cities 2 to 6, the average numbers of fires are 30, 50, 20, 15, and 25, respectively. The county wants to minimizes the average "distance to travel." Because in the county roads all run in either an east-west or a north-south direction, we use Manhattan distances rather thanEuclidean distances. For example, if the fire station were located at $(50, 40)$ and a fire occurred at city 4, the "distance to travel" is

1 / 1 point

$$|50 - 40| + |40 - 60| = 30 \text{ kilometers.}$$

We define the following parameter

$$\begin{array}{l} X_i = x\text{-coordinate of the location of city } i, i = 1, \ldots, 6, \\ Y_i = y\text{-coordinate of the location of city } i, i = 1, \ldots, 6, \\ N_i = \text{average number of fire for city } i, i = 1, \ldots, 6. \end{array}$$

Let (x, y) be the location of the hospital. A mathematical program that solves the problem is

$$\min \quad \sum_{i=1}^{6} N_i(|x - X_i| + |y - Y_i|)$$

For the above program, which of the following statement(s) is correct? Check all the correct statements.

- [ ] This program is a linear program.
- [x] This program is a nonlinear program that can be linearized.

✓ Correct

- [ ] This program is a nonlinear program that cannot be linearized.
- [x] This program has two decision variables.

✓ Correct

- [ ] This program has twelve decision variables.

4. We want to invest $1,000 to five stocks. For stock i, the current price is p_i per share, expected future price is μ_i per share, and variance is σ_i^2 per share. The objective is to maximize the total expected revenue minus one half of the variance of the revenue. Let x_i be the shares of stock i we buy. Which of the following is the correct objective function?

1 / 1 point

- ( ) $\max\ \sum_{i=1}^{5} \mu_i x_i - \frac{1}{2} \sum_{i=1}^{5} \sigma_i^2 x_i$.
- (•) $\max\ \sum_{i=1}^{5} \mu_i x_i - \frac{1}{2} \sum_{i=1}^{5} \sigma_i^2 x_i^2$.
- ( ) $\max\ \sum_{i=1}^{5} \mu_i x_i - \frac{1}{2} (\sum_{i=1}^{5} \sigma_i x_i)^2$.
- ( ) $\max\ \sum_{i=1}^{5} \mu_i x_i - (\frac{1}{2} \sum_{i=1}^{5} \sigma_i x_i)^2$.
- ( ) None of the above.

✓ Correct

5. Consider the following nonlinear program

1 / 1 point

$$\begin{array}{ll} \max & 10x_1 + 12x_2 - 20z_1 - 25z_2 - 10z_1z_2 \\ \text{s.t.} & 2x_1 + x_2 \leq 6 \\ & x_1 + 2x_2 \leq 8 \\ & x_1 \leq 3z_1 \\ & x_2 \leq 4z_2 \\ & x_1, x_2 \geq 0 \\ & z_1, z_2 \in \{0, 1\}. \end{array}$$

To linearize the above program, let $w = z_1 z_2$. After we replace $z_1 z_2$ in the objective function, which of the following constraint(s) should be added into the above program?

- ( ) $w \leq z_1$ and $w \leq z_2$.
- ( ) $w \leq z_1 + z_2$.
- ( ) $w \geq z_1$ and $w \geq z_2$.
- ( ) $w \geq z_1 + z_2$.
- (•) None of the above.

✓ Correct